2025 WSMO Tiebreaker Round

SMO Team

Tiebreaker Round Problem 1: Find the sum of the digits of $2499^2 + 2501^2$.

Tiebreaker Round Problem 2: In a regular hexagon, a beam of light is shot from one vertex of the hexagon. It bounces around inside the hexagon until it hits a vertex of the hexagon. Including the two directions along the sides of the hexagon, how many angles can the light be shot at such that it bounces at most three times before hitting a vertex?

Tiebreaker Round Problem 3: In $\triangle ABC$, D is the midpoint of $\overline{AB}$. The perpendicular bisector of $\overline{BC}$, the altitude from B to $\overline{AC}$, and line CD are concurrent. If $\angle BAC = 3\angle DCB$, the value of $\cos(\angle DCB)$ can be expressed as $\frac{\sqrt{m}+\sqrt{n}}{p}$, where m, n, and p are positive integers and m and n are not divisible by the square of any prime. Find $m+n+p$.